

Technical Architecture and Server Recommendation

Production architecture for custom commerce, research lead generation, CRM and automation

Decision Use a managed Linux virtual server in Europe, protected by Cloudflare, with the application, database and background worker deployed as a single production unit at launch. This is the right balance of cost, operational control and upgrade path for CERA's initial traffic and transaction profile. Do not use ordinary shared hosting for this workload.

Recommended launch architecture

Layer	Recommended component	Why it fits CERA
Public edge	Cloudflare Pro, DNS, CDN, WAF and rate limiting	Protects forms and checkout, serves cacheable media quickly and keeps DNS independent of hosting.
Application	Hetzner Cloud CPX32 class VM in Germany or Finland; Ubuntu LTS; 4 vCPU, 8 GB RAM, 160 GB NVMe	Enough headroom for a server-rendered website, custom API, CRM workflows and queue at launch. Current list price is about USD 35.49/month before add-ons.
Runtime	Docker Compose: web app, API, worker, Redis and reverse proxy	Repeatable deployments, isolated services and straightforward scale-up or migration.
Database	PostgreSQL on the same VM initially; encrypted daily off-server backup	A solid transactional system for orders, customers, leads, consent, audit records and inventory.
Files	Private Cloudflare R2 bucket for research briefs, attachments and backup exports; signed URLs	Keeps confidential files out of the web server and offers inexpensive, egress-free object delivery.
Email	Zoho Mail for staff mailboxes; transactional email provider for application notices	Separates staff correspondence from order, lead and system mail, improving deliverability and auditability.
Monitoring	Uptime monitor, error tracking, structured logs and alerting	A failed checkout, form, payment webhook or automation should be known quickly.

Deployment and security baseline

- Production, staging and local environments must be separated. Production credentials must never be shared in chat or committed to source code.
- Use MFA for Cloudflare, the server, source control, email, payment accounts and all CERA administrators. Each user has an individual account and role.
- Use RBAC for Owner, Ecommerce Manager, Research Sales, Customer Support, Content Editor, Scientific Reviewer and Marketing. Record critical actions in audit logs.
- Keep research attachments private; validate type, size and malware status; expose them only through authenticated, time-limited links.
- Perform encrypted daily database backup, daily media/attachment backup, retain at least 30 days and test a restore before launch and quarterly thereafter.
- Apply input validation, CSRF protection, secure cookies, rate limits, bot controls and webhook signature verification. Do not store raw card information.

Capacity and scale path

Stage	Trigger	Change
Launch	Up to normal initial commerce and research lead volume	One CPX32-class VM, Cloudflare edge, R2 attachments and automated backups.
Scale one	Sustained CPU/RAM pressure, slow database queries, or frequent background jobs	Resize compute to 16 GB RAM; move PostgreSQL to a managed service or a separate private VM; keep Redis and worker isolated.
Scale two	High order volume, campaigns or external integrations require fault isolation	Separate web/API, workers and database; introduce managed PostgreSQL, read replicas only if measured demand requires them.
No-go condition	CERA requires contractual in-country data residency or a formal availability SLA	Run a data-location and vendor review before procurement; select a provider that contractually satisfies it. Do not assume a Pakistan data centre is necessary or available.

Why this over the alternatives

Option	Recommendation	Reason
Shared cPanel hosting	Reject	It is not appropriate for custom code, queues, secure uploads, controlled deployments, CRM data or observability.
Hetzner CPX32 plus Cloudflare	Choose	Best launch cost-to-capacity. Requires competent DevOps setup and documented recovery procedure.
DigitalOcean	Acceptable alternative	Good developer experience and simple scaling, but typically higher monthly compute cost for comparable capacity.
AWS Lightsail	Acceptable alternative	Predictable bundles and AWS ecosystem, but use it only if CERA anticipates AWS services or needs a familiar enterprise procurement path.
Full AWS multi-service architecture	Defer	Unnecessary operational and financial complexity for the measured 90-day release.

Indicative monthly infrastructure cost

Service	Planning amount PKR/month	Notes
Hetzner CPX32 compute	10,700	Approx. USD 35.49 at PKR 300/USD; provider/FX charges may vary.
Server backups and off-server backup budget	2,500 - 5,000	Keep a separate restoreable backup copy.
Cloudflare Pro	6,000	About USD 20/month when billed annually;

Service	Planning amount PKR/month	Notes
		start on Free only during pre-launch if Pro features are not yet needed.
R2 storage and operations	0 - 3,000	10 GB and generous operation allowance included; cost grows with private files and media.
Transactional email, monitoring and incident alerts	2,000 - 6,000	Usage depends on volume and provider selection.
Five Zoho Mail Lite seats	1,500 - 2,000	About USD 1/user/month annually; add users as needed.
Estimated fixed operating baseline	22,700 - 32,700	Excludes payment-gateway fees, SMS/WhatsApp charges, paid advertising and support retainer.

Sources and planning notes

Provider prices are current web-published references checked in September 2026. PKR figures that depend on USD are planning estimates at PKR 300 per USD, excluding bank conversion charges, sales tax and future provider price changes.

- **CERA Medical Digital Growth Blueprint, technical architecture and security sections:** file:///home/saad/CERA_Medical_Digital_Growth_Blueprint_Stack_and_Scale.pdf
- **Hetzner Cloud price adjustment, effective 15 June 2026:** <https://docs.hetzner.com/general/infrastructure-and-availability/price-adjustment/>
- **Cloudflare plans:** <https://www.cloudflare.com/plans/>
- **Cloudflare R2 pricing:** <https://developers.cloudflare.com/r2/pricing/>
- **DigitalOcean Droplet pricing:** <https://docs.digitalocean.com/products/droplets/details/pricing/>
- **AWS Lightsail Linux bundles:** <https://docs.aws.amazon.com/lightsail/latest/userguide/amazon-lightsail-bundles.html>
- **Zoho Mail pricing:** <https://www.zoho.com/mail/zohomail-pricing.html>